

$$COP_{ref} = \frac{T_L}{T_H - T_L} \quad \text{put values in kelvin}$$

$$COP = \frac{263}{293 - 263} = \boxed{8.77}$$

So,

$$W = \frac{20.21}{8.77} = \boxed{2.304 \text{ kJ/kg}}$$



Chapter no 6

6.1 Entropy:

Entropy is the measure of randomness in a system. It helps to quantify the energy in a system that is unavailable to do work.

Entropy in thermodynamics:

$$\Delta S = \int \frac{\delta Q_{rev}}{T}$$

$\therefore \delta Q_{rev}$ = Reversible heat transfer ; T = absolute Temperature

$$\Delta S_{universe} \geq 0$$

if $\Delta S = 0 \rightarrow$ reversible

$\Delta S > 0 \rightarrow$ irreversible

$$S = S_f + x S_{fg}$$

Example 6.1

As we have

$$T_L = 0^\circ\text{C} = 273\text{K}$$

$$T_H = 60^\circ\text{C} = 333\text{K}$$

At T_L :

$$s = 1.7262 \text{ kJ/kg}\cdot\text{K}$$

and $s = s_1 = s_2$

At T_H :

$$s = 1.2857 \text{ kJ/kg}\cdot\text{K}$$

and $s = s_3 = s_4$

Now, we will use s of T_L and find pressure

By interpolation:

P	s	At 60°C
1600	1.7135	
x	1.7262	
1400	1.7360	

$$\frac{1600 - 1400}{1.7135 - 1.7360} = \frac{1600 - x}{1.7135 - 1.7262}$$

$$\frac{-888.8}{-0.0127} = \frac{-1600 - x}{-0.0127}$$

$$-8888.8 = \frac{-1600 - x}{0.0127}$$

$$-8888.8(0.0127) + 1600 = -x$$

$$x = 1487.11 \text{ kPa}$$

and

$$\text{COP} = \frac{q_H}{W_{in}} = \frac{T_H}{T_H - T_L}$$

$$\text{COP} = \frac{333}{333 - 273} = 5.55$$

Example # 6.2

$T_i = 20^\circ\text{C}$ for saturated liquid state

Now at $T = 20^\circ\text{C}$, we have

$$v_1 = 0.000923, u_1 = 87.94 \text{ kJ/kg}$$

$$s_1 = 0.3357 \text{ kJ/kg}\cdot\text{K}$$

$$m = \frac{V}{v_1} = \frac{0.001}{0.000923} = 1.083 \text{ kg}$$

$$\therefore \Delta L = 0.001 \text{ m}^3$$

Now for $T = 20^\circ\text{C}$ and $P = 400 \text{ kPa}$

$$u_2 = 276.44 \text{ kJ/kg}; s_2 = 1.2108 \text{ kJ/kg}\cdot\text{K}$$

As we know

$$\text{heat transfer} \Rightarrow Q = mT(s_2 - s_1)$$

$$Q = (1.083)(293)(1.2108 - 0.3357)$$

$$Q = 277.68 \text{ kJ}$$

Now

$$\text{Work done} = m(u_2 - u_1) + Q$$

$$W = (1.083)(276.44 - 87.94) + 277.68$$

$$W = 73.7 \text{ kJ}$$

6.6 Entropy change of a solid or liquid

For a solid or liquid under constant pressure or volume the heat transfer is related to specific heat:

$$\Delta S = \int_{T_1}^{T_2} \frac{C}{T} dt$$

where C is specific heat.

→ if C is constant:

$$\Delta S = C \ln \frac{T_2}{T_1}$$

Example # 6.3

$$T_1 = 20^\circ\text{C} = 20 + 273 = 293\text{K}$$

$$T_2 = 90^\circ\text{C} = 90 + 273 = 363\text{K}$$

$$C = 4.18 \text{ from table A.4}$$

So,

$$\Delta S = C \ln \left(\frac{T_2}{T_1} \right) = 4.18 \ln \left(\frac{363}{293} \right)$$

$$\boxed{\Delta S = 0.8954 \text{ kJ/kg-K}}$$



Problem 6.38

We have

$$T_H = 200^\circ\text{C} = 473.15\text{K}$$

$$T_L = 60.6^\circ\text{C} = 333.6\text{K}$$

$$\text{and } P_i = 20\text{kPa}$$

Now by question, we know that refrigerator works between

$$T_c = -15^\circ\text{C} = 258.15\text{K}$$

$$T_H = +20^\circ\text{C} = 293.15\text{K}$$

Now,

the heat added to Heat Engine will be:

$q_H = h_{fg}$ at $T = 200^\circ\text{C}$
because of mixed state

$$q_H = 1940.8 \text{ kJ/kg}$$

Next,

Work output of the engine:

$$\eta = 1 - \frac{T_L}{T_H}$$

$$\eta = 1 - \frac{333.6}{473.15} = 0.294$$

$$w_{\text{engine}} = \eta \cdot q_H = (0.294)(1940.8)$$

$$w = 572.41 \text{ kJ/kg}$$

COP of refrigerator

$$\text{COP} = \frac{T_c}{T_H - T_c}$$

$$= \frac{258.15}{293.15 - 258.15}$$

$$= 7.37$$

$$[\text{COP} = 7.37]$$

$$\text{As } Q_c = 1 \text{ kJ (heat loss)}$$

$$w = \frac{Q_c}{\text{COP}} = \frac{1}{7.37}$$

$$[w_{\text{ref}} = 0.135 \text{ kJ}]$$

$$\text{Now, } q_H = \frac{w_{\text{ref}}}{\eta}$$

$$q_H = \frac{0.135}{0.294}$$

$$[q_H = 0.45 \text{ kJ}] \text{ (required)}$$

